replace_experts.py

`scripts/replace_experts.py · 170 lines`

What it does

Applies the selection: loads the original model, (for the bias method) initialises bias vectors from the statistics, removes the selected experts, installs their replacement and saves the pruned model.

Run

```
uv run python scripts/replace_experts.py
```

Pipeline order: inspect_model -> build_c4_support_eval -> collect_stats -> score_experts -> replace_experts -> evaluate (before_training) -> train_biases -> evaluate (after_training) -> export_model

Reads

- the original model
- `expert_selection.json`
- `expert_stats.pt`

Writes

- `<run_dir>/replaced-model/` (full or delta checkpoint)
- `bias_initialization.json`

Notes

- Refuses stale inputs (selection made with other settings or older statistics).
- Verifies the result on a few calibration examples: bias slots return b, dropped slots return zero, rerouted tokens never reach removed experts and other tokens keep their routing.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`load_and_validate_selection(cfg, paths)`	Load expert_selection.json and check it matches the current settings and statistics.
`main()`	Script entry point: runs this stage end to end.

Settings it reads

`CHECKPOINT_FORMAT`, `MODEL_PATH or HF_MODEL_ID`, `PRUNE_EXPERTS_PER_LAYER`, `PRUNE_LAYERS`, `PRUNE_UNOBSERVED_POLICY`, `REPLACEMENT_METHOD`

Connections

Uses: `moe_pruning.data.calibration`, `moe_pruning.model.architecture`, `moe_pruning.model.instrumentation`, `moe_pruning.model.loader`, `moe_pruning.model.placement`, `moe_pruning.pruning.bias_initialization`, `moe_pruning.pruning.manifest`, `moe_pruning.pruning.replacement`, `moe_pruning.scoring.statistics`, `moe_pruning.serialization.checkpoint`, `moe_pruning.utils.cli`, `moe_pruning.utils.paths`, `moe_pruning.utils.reproducibility`